VEGA

version 1.2.0 - BETA

USER'S GUIDE

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1. Introduction

VEGA is an application that allows to apply several QSAR models to the given chemicals provided by the user, returning as output complete reports (in plain text and in PDF format) containing the predictions and all needed supporting information. It is a JAVA standalone application and it does not require any network communication (chemicals are processed on the local machine, no information is shared across the Internet).

Currently, there are **95 models** implemented in VEGA:

1	Mutagenicity (Ames test) model (CAESAR) - ver. 2.1.13
2	Mutagenicity (Ames test) model (ISS) - ver. 1.0.3
3	Mutagenicity (Ames test) model (SarPy-IRFMN) - ver. 1.0.8
4	Mutagenicity (Ames test) model (KNN-Read-Across) - ver. 1.0.1
5	Mutagenicity (Ames test) CONSENSUS model - ver. 1.0.4
6	Developmental Toxicity model (CAESAR) - ver. 2.1.8
7	Developmental/Reproductive Toxicity library (PG) - ver. 1.1.2
8	Carcinogenicity model (CAESAR) - ver. 2.1.10
9	Carcinogenicity model (ISS) - ver. 1.0.3
10	Carcinogenicity model (IRFMN-ISSCAN-CGX) - ver. 1.0.1
11	Carcinogenicity model (IRFMN-Antares) - ver. 1.0.1
12	Carcinogenicity oral classification model (IRFMN) - ver. 1.0.1
13	Carcinogenicity oral Slope Factor model (IRFMN) - ver. 1.0.1
14	Carcinogenicity inhalation classification model (IRFMN) - ver. 1.0.1
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16	Carcinogenicity in male rat (CORAL) - ver. 1.0.0
17	Carcinogenicity in female Rat (CORAL) - ver. 1.0.0
18	Acute Toxicity (LD50) model (KNN) - ver. 1.0.0
19	Skin Sensitization model (CAESAR) - ver. 2.1.7
20	Skin Sensitization model (IRFMN-JRC) - ver. 1.0.1
21	Skin Sensitization model (NCSTOX) - ver. 1.0.1
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39	BCF model (Arnot-Gobas) - ver. 1.0.1
40	BCF model (KNN-Read-Across) - ver. 1.1.1
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42	Fish Acute (LC50) Toxicity model (NIC) - ver. 1.0.1
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The current release of the application is a **beta version**, please note that some of the guides for the most recent models are not available yet, and will be included in the next release of VEGA.

Please refer to the VEGA website for the list of available model’s guides and QMRF.

2. Installation

VEGA is a JAVA application that works on every operating system (Windows/Linux/Mac) supporting Java. **Java 11** or higher is required. You can download OpenJDK installers from https://jdk.java.net/ or follow the instructions available on the OpenJDK website (https://openjdk.java.net/).

Download and unpack the zipped file. To start the application, just move to the VEGA folder and run the file *Vega-launcher-WIN.bat* (on Windows platforms) or *Vega-launcher-LINUX.sh* (on Linux platforms).

If you are using a Windows system with an older version of Java note that you can still run the application using the *Vega-launcher-WIN.bat* launcher, as it uses a local distribution of Java included in the folder.

If you experience problems with the above mentioned launchers, you can just run the file *Vega-GUI-1.2.0.jar* with the Java Runtime Environment (JRE). On most systems, it is enough to double-click it. If you are not able to directly run it , open a command line window (like Command Prompt on Windows systems or BASH shell on Linux), move to the Vega folder and type:

java -jar Vega-GUI-1.2.0.jar

Note: Vega may encounter problems when processing large datasets and/or using a high number of models at the same time. You can modify and use the launcher files (*Vega-launcher-WIN.bat* and *Vega-launcher-LINUX.sh*) to assign more memory to the application. In the launcher file modify the parameter "-Xmx1024m", where 1024m means that Vega uses 1,024 MB (1 GB), setting a higher number (for instance -Xmx2048m for running Vega with 2 GB of memory).

3. How to Use

VEGA has a user-friendly interface, organized in three tabs corresponding to the steps needed to perform calculations. In order to run the models, all the parameters of the three tabs must be correctly set. In the first tab,INSERT, you have to build the compounds dataset. In the second tab, SELECT, you have to select the models to be calculated. In the third tab, EXPORT, you have to select the type and the destination folder for saving the generated reports.

3.1 Insert

In this tab, you have to insert one or more structure(s) in order to build the dataset that will be used for the calculations. You can insert the structure by directly typing it, using the SMILES format, in the "Load" text field; it is also possible to copy-and-paste single SMILES in the text field. Then, click the "Load" button, or just press enter to insert the structure. If the SMILES format of the structure is correctly typed, then it is imported in the table below the text field.

Also, you can import one or more structure(s) from SDF or SMILES file formats. SDF (structure-data file) is a chemical data file format developed by MDL. This kind of text-based file is specifically formatted to include multiple structures supplied with data (e.g., name, CAS, molecular mass, activity etc.). SMILES files are text files containing multiple SMILES strings (one molecule for each line of the file). For this kind of files, also tab-separated text files are accepted: for each line, other information for the given SMILES molecule (like its id or CAS number) can be reported, separated by a tab character.

To import a file, click the "Import file" button and select the file to be loaded. Depending on the type of the file, further information could be required:
- For SDF files, if some tags are available a form will give you the chance to choose one of the tags as the id (name) of the molecules that will be loaded.
- For SMILES files, if the file contains tab-separated fields a form will give you the chance to choose what field contains the SMILES string, and to optionally choose another field as the id (name) of the molecules that will be loaded.
In both cases, it will be possible to select the option for parsing the chosen molecule's id as a CAS number: the read string will be normalized so that CAS numbers written with separation characters other than "-", or CAS written without any separation character will be reported in a default style (like 12345-67-8).

After successfully importing the structures using one or both of the ways above, you can visualize the structure by clicking on it.

By selecting some of the imported structure(s) and clicking the "Delete" button it is possible to erase them. To clean the whole table just click "Delete All" button.

3.2 Select

In the second tab, you have to select the models to be calculated. The currently available models are organized in six main categories that can be filtered:

- Human Toxicity
- EcoToxicity
- Fate & Distribution
- Physical-Chemical properties
- Human PBPK
- Ecological PBPK

For each category, all models available are shown grouped by end-point.

By clicking the question mark icon near each model, you can access to the details of the model and to the list of available files for that model: its user's guide (PDF document); the original dataset used for building the model (plain text file) containing the CAS number, the SMILES, the set (training or test set) and the experimental value for each compound; the QMRF document if available.

3.3 Export

In the third tab, you have to select the type of the reports to be generated, and the destination folder where the reports will be saved. Reports can be saved in two different formats, PDF and plain text. The supported output choices are the below:

- Multiple PDF documents – Results are saved in PDF format, one file for each calculated model
- PDF single document (ordered by model) – Results are saved in a single PDF report, ordered by model
- PDF single document (ordered by molecule) – Results are saved in a single PDF report, ordered by molecule

- Summary as tab-separated text file – Results are saved in a single text file, where for each model is reported only the final assessment
- Multiple tab-separated text files – Results are saved in text files, one for each model

For each choice, you have also to select the destination folder where the reports will be saved after the calculation. It is possible to select one or more export choices.

3.4 Predict

When the parameters of the three tabs described above are correctly filled, just click “Predict” button to start calculations. Please note that the calculation and report generation task may take long time, depending on the number of the imported substances and the selected models. In particular, note that the task of creating PDF reports can be particularly time-consuming when calculation are done for high numbers of compounds

4. Acknowledgements and contacts

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Please check each model's guide for acknowledgements on its development.

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